Fraud Shield Banking Analytics Summary

Objective

The objective of this project was to analyse banking transaction data using Power BI and create an interactive dashboard to monitor transaction performance, customer behaviour, merchant performance, and potential fraud activities.

Data Preparation and Transformation

The cleaned dataset was imported into Power BI. The following data preparation steps were performed:

- Corrected data types for Transaction Date, Transaction Time, Transaction Amount, Customer ID, Merchant ID, and IP Address.
- Removed errors and validated data quality.
- Created dimension tables for Customers and Merchants to build a star schema model.
- Established relationships between fact and dimension tables.

Data Modelling

A star schema model was implemented consisting of:

- Fact Transactions Table
- Customer Dimension Table
- Merchant Dimension Table

Relationships were created using Customer_ID and Merchant_ID to improve reporting performance and data organization.

DAX Measures Created

Several measures were developed to support business analysis:

- Total Transactions
- Total Transaction Amount (Million)
- Distinct Customers
- Distinct Merchants
- Transactions per Customer
- Average Revenue per Merchant

- Fraud Transactions
- Fraud Rate (%)
- High Distance Transactions

Executive Overview Dashboard

This dashboard provides a high-level view of banking operations through:

- KPI Cards showing Total Transactions, Transaction Amount, Customers, Fraud Transactions, and Maximum Transaction Amount.
- Transaction Trend Analysis using a Line chart.
- Revenue Analysis by Merchant Category using Stacked Column Chart.
- Transaction Type Distribution Analysis using Pie Chart.
- Fraud vs Non-Fraud Transaction Analysis using Donut Chart.

Customer & Merchant Analysis Dashboard

This dashboard focuses on customer and merchant behaviour:

- KPI Cards showing Customer, Merchants, Transaction per Customer and Average Revenue per Merchant.
- Customers by Transaction Amount using Clustered Bar Chart.
- Merchants by Revenue using Clustered Column Chart.
- Merchant Category Performance using a Tree map.
- Customer Home & Merchant Analysis using a map visual.
- Card Type Distribution using a Donut Chart.
- Transactions by Merchant Category using a Stacked Column Chart.

Fraud Analysis Dashboard

A dedicated dashboard was created to identify suspicious activities:

- KPI Cards showing High Distance Transactions, Fraud Transactions, Fraud Rate.
- Fraud vs Non-Fraud Transaction Distribution using a Donut Chart.
- Fraud by Transaction Type using a Clustered Column Chart.
- Fraud by Merchant Category using a Stacked Bar Chart.

- Scatter analysis of Transaction Amount versus Distance from Home using a Scatter Chart.

Key Insights

The dashboard enables stakeholders to:

- Monitor overall transaction performance.
- Identify high-value customers and merchants.
- Analyse customer geographic distribution.
- Detect unusual transaction patterns.
- Track fraud-related activities and risk indicators.
- Support data-driven decision-making for banking operations and fraud prevention.

Conclusion

The Fraud Shield Banking Analytics Dashboard successfully transforms raw transaction data into meaningful business insights through data cleaning, modelling, DAX calculations, and interactive visualizations. The solution helps monitor operational performance while supporting fraud detection and risk analysis.